

WRESTLING EMPIRE

Canonical Data Schemas & Domain Contracts

Phase 0 | Version 0.1

Product authority: Game Design Document v0.2

Engineering authority: Technical Architecture & Vertical Slice Specification v0.2

Execution gate: Development & Build Plan v0.1, Phase 0

Status: PHASE 0 COMPLETE / READY FOR PHASE A-C IMPLEMENTATION

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0. Document Control and Authority

This artifact is the code-adjacent source of truth required by the Phase 0 build plan. It translates the accepted design and architecture into exact serialization, identity, reference, command, persistence, Chronicle, hidden-information, narrative-context, campaign/map, and fixture contracts. It does not reopen gameplay design and it does not manufacture tuning values that the governing documents intentionally leave open.

Authority	Role in this artifact	Controlling rule
Game Design Document v0.2	Product authority	Defines what the game is, its locked design constitution, entity responsibilities, hidden information, history, kayfabe, map/influence model, and slice intent.
Technical Architecture & Vertical Slice Specification v0.2	Engineering authority	Defines system ownership, domain boundaries, headless simulation, IDs, commands/events, RNG, persistence, Chronicle, knowledge, NarrativeContext, and architecture decisions.
Development & Build Plan v0.1	Execution gate	Defines Phase 0 work packages, deliverables, rework triggers, and exit criteria.

Precedence. Where this artifact makes an implementation-level choice not fixed by the authorities, the choice is deliberately conservative and reversible. A contradiction with a locked design/architecture rule is a defect. A genuinely unresolved product or evidence-dependent question is recorded in the Open-Field Ledger rather than silently guessed.

0.1 Phase 0 Completion Statement

- All slice-critical identity families and reference semantics are defined.
- Every slice-critical domain family has a field-level machine-readable schema or an explicit definition-backed representation.
- Static authored content is separated from mutable runtime state.
- Player, AI, automation, and system mutations share one command/result path.
- Simulation randomness has one controlled service contract and persisted boundary state.
- Save, Chronicle, migration, atomic-write, and recovery boundaries are defined.
- Knowledge projections prevent normal UI from obtaining hidden truth directly.
- NarrativeContext is machine-validatable across authored content and presentation surfaces.
- Campaign/map/mod content is data-driven and free of Great Lakes/1975/count ceilings.
- Valid and deliberately invalid fixtures are included and machine-validated.
- Open questions are recorded without blocking Phase A-C.

1. Identity and Reference Contract (Work Package 0.1)

1.1 ID families

ID family	Canonical form	Examples	Semantics
Static content ID	namespaced lowercase dotted ID	market.alpha; role.wrestler; tech.media.local_tv	Immutable definition identity inside ContentRegistry. Used for maps, markets, regions, venues, rulesets, roles, skills, media technologies, narrative keys and other authored definitions.

ID family	Canonical form	Examples	Semantics
Runtime entity ID	type:opaque-token	person:PER00001; promotion:PRO00001; contract:CON00001	Opaque identity unique for campaign lifetime. Token has no gameplay semantics and must never be parsed for date/order/geography.
Definition-backed world state	content ID is runtime key	market.alpha; region.example; venue.alpha_hall	Markets/regions/venues are singleton overlays on static world definitions, so a second spawned runtime identity would duplicate truth.
Localization key	static content-ID grammar	event.departing_champion.offer. body	Stable externalized prose key. Narrative context is metadata separate from prose.
Schema version	positive integer	state_schema_version: 1	Versioned independently where migration boundaries require it.

1.2 Non-reuse, lifetime, tombstones, and identity epochs

- Runtime IDs are never reused inside a campaign, including after retirement, death, closure, disbandment, merger, title retirement, or contract expiry.
- Required current-state references must resolve when state loads or validates. Optional references are explicit nulls, not magic empty strings or missing keys.
- Retirement and death do not create a new Person. Role changes never create a new Person. Person identity survives career position changes.
- Promotion closure or merger may remove an organization from hot mutable state after compaction, but Chronicle must retain a tombstone/identity record. A merger records successor/survivor references rather than redirecting old IDs invisibly.
- Relocation does not terminate identity. Home-region, name, ring identity, logo, ownership, and brand changes are dated epochs/events. Historical presentation uses the identity valid at the requested date.
- Current-state compaction is legal only after all historical references can resolve through active state or Chronicle identity catalog.
- Serialized references are IDs only. Node, Control, NodePath, Resource instance IDs, scene addresses, UI selections, caches, and object pointers are never authoritative.

1.3 Reference validation rules

Rule	Failure code	Required behavior
Malformed ID	ID001	Reject file/state with exact path.
Duplicate ID	ID002	Reject duplicate/conflicting definition or runtime identity.
Missing required reference	REF001	Reject load/command before mutation.
Collection key differs from contained canonical ID	REF002	Reject as ambiguous identity.
Wrong reference family/type	REF003	Reject where target family is known.
Historical ID unresolved by active state or tombstone	CHR002	Reject Chronicle integrity validation.

2. Core State Contracts (Work Package 0.2)

CampaignState is the authoritative mutable state for one save. ChronicleStore is its authoritative historical companion and is not nested into the hot mutable graph. ContentRegistry is immutable for the session. Derived values, caches, UI sorting, presentation state, animation state, and scene references are excluded from authoritative state.

```

CampaignSession
  current_state: CampaignState
  chronicle: ChronicleStore
  content: ContentRegistry (immutable)
  services: CommandRouter / QueryService / RandomService / SaveService

```

2.1 Canonical CampaignState fields

Field	Type	Presence	Contract
state_schema_version	integer >= 1	required	Current-state migration boundary.
campaign_pack_id	ContentId	required	Campaign definition identity.
campaign_pack_version	SemVer	required	Pack version used by this save.
content_fingerprint	SHA-256	required	Resolved content identity for compatibility/recovery.
ruleset_id	ContentId	required	Ruleset/tuning reference; formulas are not hard-coded into state.
current_date	YYYY-MM-DD	required	Current in-game date.
turn_number	integer >= 0	required	Monotonic strategic-turn count.
ownership_seat	OwnershipSeatState	required	Current player promotion/owner seat without inventing succession mechanics.
world_state	object	required	Active era, available technologies, global wrestling-interest state.
people/promotions/...	ID-keyed dictionaries	required	Authoritative entity stores; keys must equal canonical entity identity.
knowledge_bases	PromotionId -> KnowledgeBase	required	Organization-specific observations. Player and AI may use same model.
event_state	EventState	required	Authored-event cooldown/firing/choice state.
rng_state	RandomState	required	Controlled random boundary state.
victory_state	VictoryState	required	Open-ended campaign/end-state status; thresholds remain ruleset data.

2.2 Slice-critical entity ownership/reference matrix

Entity	Identity	Owns authoritative truth	References / derived elsewhere
PersonState	runtime PersonId	Human identity, skills/traits/potential, roles, health, career, local audience state, contract refs.	Roster position and staff function are role/contract state, not species. Hidden truth reaches UI only through knowledge projection.
PromotionState	runtime PromotionId	Organization identity, ownership, money, prestige, morale, momentum, strategic refs.	Market influence values live on MarketState. Roster/staff are derived from contracts/role assignments rather than duplicated lists.
MarketState	market ContentId	Current interest, economic strength, influence component vectors, hot/cold/effects.	Static population, coordinates, culture, venue stock and infrastructure definitions live in content. Composite influence is derived by ruleset.
Region	region ContentId	No invented independent v0.1 economic truth. RegionState only carries region-scale effects/hot state extension point.	Membership, culture/economy profile and geometry are RegionDefinition content; normal aggregates are derived from markets.
TouringCompanyState	runtime TouringId	Promotion membership, scoped person assignments, carried titles, route, directives, budget, fatigue/cohesion/hot state.	Does not own people or titles.

Entity	Identity	Owns authoritative truth	References / derived elsewhere
ContractState	runtime ContractId	Person-promotion relationship, term, compensation, exclusivity, clauses, leverage/dispute state.	Person/Promotion carry reference lists for indexing; contract is relationship truth.
ChampionshipState	runtime ChampionshipId	Current title status, prestige, holder(s), recognition.	Lineage is Chronicle history, not an ever-growing live array.
ProgramState	runtime ProgramId	Participants/sides, purpose, phase, heat, momentum, objective/direction, chemistry truth.	Player-visible chemistry may be filtered through knowledge/reporting.
MediaDealState	runtime MediaDealId	Specific promotion-medium-outlet term, reach, money, exclusivity/requirements/restrictions.	Medium/technology behavior is static/ruleset content. Local TV is one implementation.
VenueState	venue ContentId	Current availability, promotion relationships, active restrictions.	Capacity/base cost/prestige/market are static VenueDefinition.
AgreementState	runtime AgreementId	Promotion parties, status, dated clauses, trust effect and violations.	Formal territory is a clause and can be violated; markets never gain an owner field.
RelationshipState	runtime RelationshipId	Person-person relationship strength/trust/grievance/tags.	Player knowledge about the relationship is separate observation data.

2.3 Numeric representation policy

The governing GDD explicitly leaves exact numerical tuning open. Phase 0 therefore fixes representation without inventing game balance. Bounded perception/state quantities use normalized [0,1] scalars; bidirectional quantities such as momentum, trust, relationship strength and trait axes use [-1,1]. Money is integer minor currency units plus a currency definition ID. These ranges are serialization contracts, not statements about formulas, decay rates, thresholds, victory requirements, or optimal values.

3. Static Content vs Runtime State (Work Package 0.3)

Concern	Static / immutable ContentRegistry	Mutable CampaignState / Chronicle
Geography	MapDefinition, MarketDefinition, RegionDefinition, coordinates, connections, culture/infrastructure tags.	Market interest/economy, influence, venue availability/relationships, temporary effects, historical projections.
People	Person seed/template, role/skill/trait definitions.	Person identity/career/skills/traits/roles/health/contracts/audience state.
Promotions	Promotion seed, strategy profile, style/tuning definitions.	Actual organization identity, ownership, cash, prestige, roster relationships, deals, programs, titles.
Media/era	MediaTechDefinition, medium/outlet definitions, availability/rules, EraDefinition.	Adopted technologies, specific MediaDealState, current world state.
Events/narrative	EventDefinition conditions/choice/effect refs, localization keys, NarrativeContext, content tags.	Firing/cooldown/choice history and emitted DomainEvents.
Rules/balance	Ruleset and tuning profiles.	Only resulting current values and durable facts; formulas are not serialized as campaign truth.

Hard-coding audit rule: a schema, validator, command, or engine branch that depends on a city name, Great Lakes, 1975, exactly three promotions, exactly 12-15 markets, or a fixed wrestler count is a Phase 0 defect.

4. Command, Result, and Domain Event Contracts (Work Package 0.4)

4.1 CommandEnvelope

Field	Contract
command_schema_version	Positive integer command-envelope schema version.
command_id	Stable runtime command ID.
command_type	Namespaced command type such as command.set_route.
issued_for_turn / issued_on	Optimistic turn/date context used during validation.
issuer	player / ai / automation / system, with optional promotion/person scope.
ordering	Turn phase, integer priority, issuer_key. Command ID is final tiebreaker.
payload	Command-specific data validated by command handler schema/validator. Unknown command types are rejected.

Deterministic queue order is the tuple (turn phase ordinal, priority, issuer_key, command_id). Every deterministic collection is iterated by stable canonical ID. AI planners issue the same application/domain commands as the player where practical and receive the same validation results.

4.1.1 Typed v0.1 command payload catalog

Family	Command types	Payload contract
Organization	hire_staff; assign_role; fire_person; set_booker; adjust_budget	Exact person/promotion/role/budget references and terms; no UI-owned mutation fields.
Touring	create_touring_company; assign_person; set_route; set_directive; split_company; merge_company	Company IDs, scoped assignments, typed TourStop routes, directives and budgets.
Creative	start_program; end_program; set_champion; approve_major_outcome; push_person; protect_person	Program sides/objectives, championship holders, bounded owner directives/decisions.
Contracts	offer_contract; counter_offer; renew_contract; release_person	Person/promotion/contract IDs plus shared compensation, exclusivity and clause shapes.
Markets / media	book_market_focus; set_local_media_spend; sign_media_deal	Market/medium/outlet/reach references and Money terms.
Diplomacy	propose_agreement; violate_territory; accept_talent_share	Promotion parties, AgreementClause data, market and talent references.
Campaign	advance_month; save_campaign	Empty advance payload; save display-name request only.

The machine schema exposes these through TypedCommand: command_type is a constant for each variant and payload is closed to that command-specific schema. New commands therefore require an explicit schema addition rather than a screen inventing a private payload shape.

4.2 CommandResult and validation errors

Results are structured and localization-ready. Rejections happen before mutation. Error records contain a stable code, data path, localization key, and structured details. Presentation never needs to parse English error prose to determine behavior.

4.3 DomainEvent

DomainEvents are facts emitted by simulation systems into TurnResult. A transient event may lack permanent identity. ChronicleCommitter assigns a permanent event ID and monotonic sequence when the event is

journalled. Events may carry machine-readable facts, causal factors, caused-by links, historical class, optional narrative key, and optional NarrativeContext. Events do not replace metric series or historical projections.

Representative event family	Examples from governing architecture
Shows	domain_event.show_completed
People	domain_event.person_injured
Championships	domain_event.championship_changed
Markets	domain_event.market_influence_changed
Contracts	domain_event.contract_expired
Diplomacy	domain_event.territory_violated
Hot/cold	domain_event.promotion_became_hot
History/records	domain_event.historical_record_broken

5. Randomness Contract (Work Package 0.5)

- All simulation randomness flows through RandomService. No domain subsystem calls Godot/global random functions or wall-clock functions directly.
- v0.1 uses one root RNG stream behind the RandomService seam. Calls are diagnostic-tagged (for example injury, scouting_error, crowd_variance) but tags do not create independent hidden streams.
- CampaignState stores provider ID, seed, internal state, captured turn, and stream policy at turn boundaries. SaveManifest records the checkpoint.
- The pinned Godot engine line defines reproducibility expectations for the initial wrapper. A project-owned PRNG remains an open engineering question, not a Phase 0 assumption.
- Historical Replay never depends on re-running RNG. Recorded Chronicle history is authoritative.

6. Save and Migration Contracts (Work Package 0.6)

```

user://saves/<save_id>/
  manifest.json
  state.<codec>
  chronicle/
    manifest.json
    events/ metrics/ deltas/ checkpoints/ indexes/
    artifacts.<codec>
    identity_catalog.<codec>
  thumbnail.webp (optional)
  backups/

```

6.1 SaveManifest required fields

Family	Required information
Versions	save_schema_version, architecture_version, game_version, pinned engine major/minor, chronicle_schema_version.
Content	campaign pack ID/version/hash, enabled pack IDs/versions/hashes, resolved content fingerprint.
Identity/time	save ID, display name, created/updated UTC timestamps, in-game date.
Player seat	player promotion ID and OwnershipSeatState.
RNG	provider/seed/state/turn checkpoint metadata.
Chronicle	head date/sequence, checkpoint generation, Chronicle integrity summary, Chronicle manifest path.
Integrity/recovery	state path, optional integrity checksum and previous-backup pointer.

6.2 Serialization, atomicity, recovery

- Explicit codecs serialize primitive/data structures and stable IDs. Reflection-only magic serialization is not authoritative.
- Unknown future fields are not silently accepted in schema v1. Older binaries refuse unsupported newer schemas unless a documented forward-compatibility strategy is added later.
- Required references are load errors; optional references are explicit null. Numbers are finite and range-validated.
- Content definitions are referenced, not duplicated wholesale into saves.
- Committed Chronicle records are append-oriented and immutable during ordinary play.
- Writes use temporary paths and publish the new state/Chronicle head only after validation. The last known-good save survives failed writes.
- Android background/pause requests a safe checkpoint; the system never assumes graceful shutdown.

6.3 Migration transaction

State and Chronicle schemas may version independently but migrate as one campaign transaction: create migration backup; transform sequentially $vN \rightarrow vN+1$ in memory/temporary storage; rebuild non-authoritative indexes when required; validate current-state and historical invariants; write and close temporary output; atomically replace target. A migration never overwrites the only copy and never silently rewrites the meaning of committed historical facts.

7. Campaign Chronicle Contracts (Work Package 0.7)

ChronicleStore is the persistent historical companion to CampaignState. It is append-oriented, stable-ID based, queryable by date/entity/type, and logically separable from hot state. Events answer what happened. Metric series answer how selected values moved. HistoricalProjection answers what the relevant wrestling world looked like at a date.

Component	Canonical responsibility	Authoritative?
EventJournal	Journaled DomainEvents with permanent event/sequence IDs, dates, entity refs, facts, causes and narrative metadata.	Yes
MetricSeriesStore	Ordered HistoricalMetricDefinition-driven samples.	Yes
CheckpointStore	Periodic HistoricalProjection keyframes containing reconstruction fields only.	Yes
DeltaStore	Ordered add/replace/remove changes between checkpoints.	Yes
ChronicleIndex	Date/entity/event/record and lifetime lookup indexes.	Rebuildable where possible
ArtifactIndex	Metadata references to historical ephemera and their original NarrativeContext.	Metadata authoritative; large media external
Identity catalog / tombstones	Historical resolvability and dated identity/presentation epochs after entities leave active state.	Yes
ChronicleQueryService	Read-only reconstruction/query facade. Never mutates CampaignState.	Service, not stored truth

7.1 HistoricalProjection minimum contract

- Date and schema version.

- Active/inactive promotion identity, owner, roster, prestige and momentum summaries.
- Championship status, holders and prestige.
- Market wrestling interest, influence component vectors and relevant hot/cold state.
- Formal agreement status/parties/clauses.
- Media footprint/reach summaries.
- Identity references needed to render historical entity presentation.

7.2 Reconstruction rule

get_projection(date) locates the nearest checkpoint at or before the target, copies its HistoricalProjection, then applies ChronicleDelta operations in sequence through the requested date. Zero-change deltas are omitted. Checkpoint cadence and physical compression are policy/codec parameters, not schema assumptions. Historical Replay, historical maps, timelines, graphs, and archive browsing read this recorded state and never invoke simulation mutation or old AI/RNG logic.

8. Knowledge and Narrative Boundaries (Work Package 0.8)

8.1 KnowledgeBase / KnowledgeProjection

KnowledgeBase stores structured observations owned by a promotion. Each observation identifies a subject and field/domain, estimate form (unknown/qualitative/range/exact), confidence, source, date, bias and error margin. Better familiarity/scouting may narrow uncertainty. The same model can be used by player and AI organizations; any deliberate AI information advantage must be explicit outside this contract.

- Normal UI QueryService returns player-specific KnowledgeProjection for hidden domains. It does not hand screens a PersonState/PromotionState truth object and ask the screen to remember which fields are secret.
- Exact skills, hidden traits/potential, injury risk, contract intentions, rival finances/strategy and undiscovered relationships remain authoritative simulation truth but are not automatically exposed.
- Debug/simulation-inspector tooling may show truth and projection side-by-side but is not a normal player surface.
- KnowledgeProjection schema has no true_value escape hatch. Unknown extra fields are rejected.

8.2 NarrativeContext

Context	Permitted use	Core prohibition
public_kayfabe	Broadcasts, commentary, interviews, newspapers, fans, posters and in-world public artifacts.	Must not acknowledge scripted outcomes, fictional personas, audience-winking meta commentary, or game mechanics.
backstage_business	Owner-facing booking, contracts, agents, creative and management surfaces.	May use legitimate wrestling-business language but does not become meta commentary about the game itself.
system_neutral	Settings, accessibility, technical notices and non-character system language.	Not a loophole for public narrative to break kayfabe.

Every narrative-bearing definition declares exactly one context. Every narrative surface declares allowed contexts. Localization inherits source context. Context mismatch is a validator error. Archive artifacts retain the context under which they were created.

9. Campaign and Map Schema Contract (Work Package 0.9)

9.1 CampaignPackManifest

Field family	Contract
Identity/version	pack_id, positive schema_version, semantic pack_version.
Dependencies	Explicit dependency/version specs resolved deterministically.
Scenario refs	map_id, start_date, ruleset_id. Start year is content, never code.
Files	Logical content-family -> safe relative path. Absolute paths, URI/executable references and traversal are forbidden.
Localization	One or more locale identifiers.

9.2 Map topology

Simulation topology is a graph. A MapDefinition contains market IDs, region IDs, travel connections and media-adjacency rule references. Visual coordinates, route polylines, region polygons and art are optional metadata layered on top. Travel edges carry modes, base distance/time/cost index and availability windows. Media reach is not required to mimic physical travel edges.

9.3 Data-only mod safety

- Initial mods may add/override validated data and assets but cannot execute arbitrary scripts.
- Overrides require explicit target IDs and deterministic load order. Duplicate/conflicting definitions fail loudly.
- Pack file references must be safe relative paths inside allowed roots. Path traversal, remote URIs and executable references are invalid.
- Unknown fields are rejected by schema v1 rather than silently interpreted.
- Save manifests record exact enabled pack identities/versions/hashes. Missing or incompatible packs produce recovery/compatibility failure, never silent substitution.

10. Contract Fixtures (Work Package 0.10)

The package contains valid and invalid JSON payloads intended for Phase B/C validators and regression tests. The fixtures use deliberately generic Alpha/Beta geography so a passing test cannot accidentally rely on Great Lakes, Detroit, 1975-specific engine branches, or a fixed promotion count. The 1975 date appearing in a fixture is content only and is not a schema constraint.

Fixture	Expectation	Purpose
valid/campaign_bundle.json	PASS	Cross-family state/content shape with two markets, one promotion, people, contract, title, touring company, local-TV deal, knowledge and RNG.
valid/command_set_route.json	PASS	Player command through canonical envelope.
valid/journaled_domain_event.json	PASS	Permanent Chronicle event identity and sequence.
valid/knowledge_projection.json	PASS	Hidden-information-safe observation projection.
valid/chronicle_bundle.json	PASS	Event/metric/checkpoint/index/artifact/tombstone boundaries.
valid/save_manifest.json	PASS	Version/content/RNG/Chronicle-aware save manifest.
invalid/bad_runtime_id.json	FAIL	Malformed runtime ID.
invalid/broken_required_reference.json	FAIL	Missing required person reference.

Fixture	Expectation	Purpose
invalid/hidden_information_leak.json	FAIL	Knowledge projection attempts to expose true_value.
invalid/narrative_context_mismatch.json	FAIL	Backstage content sent to public-only surface.
invalid/unsafe_pack_path.json	FAIL	Path traversal attempt.
invalid/collection_key_id_mismatch.json	FAIL	Dictionary key conflicts with entity canonical ID.
invalid/duplicate_active_assignment.json	FAIL	Duplicate active role assignment.
invalid/journaled_event_missing_identity.json	FAIL	Journaled event lacks permanent ID/sequence.
invalid/unknown_command_type.json	FAIL	Command type has no typed payload contract.

11. Machine-Consumable Schema Package

schemas/we.phase0.schema.json is the canonical JSON Schema Draft 2020-12 definition set. Per-family wrapper schemas provide convenient entry points. JSON Schema handles structural validation; the included semantic validator adds graph/reference and cross-field invariants that JSON Schema alone cannot reliably express.

Schema family	Primary definitions
Common/value objects	ContentId, RuntimeId, Date, SemVer, Money, normalized scalars, HotColdState, Explanation, IdentityEpoch.
Current state	CampaignState plus all slice-critical state entities and supporting Relationship/Role/Health/Audience values.
Application	CommandEnvelope, TypedCommand and 29 command-specific payload schemas, CommandResult, ValidationError, DomainEvent, TurnResult.
Randomness	RandomState.
Knowledge/narrative	KnowledgeBase, KnowledgeObservation, KnowledgeProjection, NarrativeContext, NarrativeSurfacePayload.
Chronicle	ChronicleManifest/Bundle, DomainEvent journal records, metrics, HistoricalProjection, checkpoints, deltas, indexes, artifacts, tombstones.
Persistence	SaveManifest and pack/version refs.
Static content	CampaignPackManifest; Map/Market/Region/Venue; Person/Promotion seeds; Event/MediaTech; Era/Ruleset; Role/attribute/title/media-outlet definitions.

12. Open-Field Ledger

These questions are intentionally unresolved because the governing documents require evidence, profiling, later product design, or playtesting. They do not block Phase A-C. The current contract preserves the required extension point without pretending the answer is known.

OF-001 - Exact promotion-influence component weights and whether infrastructure is player-visible.

Source: Technical Architecture v0.2 §17 Q2; ADR-006 tentative

Current contract: Store audience/media/business/infrastructure normalized components; composite is derived by ruleset, not persisted as truth.

Resolution evidence: Phase D/E simulation tuning and map readability tests.

OF-002 - Exact audience-state geographic granularity beyond per-market slice data.

Source: Technical Architecture v0.2 §17 Q3

Current contract: PersonState supports sparse audience_by_market. Region/global aggregation remains derived/extension-point behavior.

Resolution evidence: Long-run scale profiling and full-map design.

OF-003 - Which Chronicle metrics sample monthly versus quarterly/annual/material-change.

Source: GDD v0.2 §31 Q15; Technical Architecture v0.2 §17 Q4

Current contract: HistoricalMetricDefinition externalizes cadence, aggregation, precision and retention; no cadence is hard-coded in schemas.

Resolution evidence: Phase D/H storage and usefulness measurements.

OF-004 - HistoricalProjection checkpoint cadence.

Source: Technical Architecture v0.2 ADR-020; §17 Q5

Current contract: Checkpoint generation/cadence is metadata/policy. Annual checkpoints + monthly deltas are the tentative initial implementation, never a schema assumption.

Resolution evidence: Chronicle size and reconstruction-latency profiling.

OF-005 - Physical Chronicle persistence codec/chunking/compression format.

Source: Technical Architecture v0.2 §8.1, §8.6, §17 Q6

Current contract: Logical stores remain separate behind ChronicleCodec; JSON/coarse segments are permitted initially.

Resolution evidence: Measured save size, write cost and Android I/O behavior.

OF-006 - Whether to replace the pinned Godot RNG wrapper with a project-owned PRNG.

Source: Technical Architecture v0.2 §6.4; §17 Q7

Current contract: RandomService single entry point; provider/state recorded at turn boundaries; Historical Replay does not depend on RNG replay.

Resolution evidence: Cross-engine diagnostic-repeatability need.

OF-007 - Hidden-information policy while inspecting active-campaign historical dates.

Source: Technical Architecture v0.2 §9.7 and §17 Q10

Current contract: Chronicle stores authoritative history; normal live QueryService never bypasses KnowledgeBase. Historical presentation policy remains separate.

Resolution evidence: Historical UI/product decision before polished replay.

OF-008 - Archive artifact permanence versus regenerable cache by artifact class.

Source: Technical Architecture v0.2 §17 Q12

Current contract: ArtifactIndex stores metadata/reference only; no permanence field is made authoritative yet.

Resolution evidence: Artifact production and storage-cost evidence.

OF-009 - Final map geometry import/storage pipeline.

Source: Technical Architecture v0.2 §17 Q13

Current contract: Map topology is authoritative; optional map-space coordinates/polylines/polygons are content metadata.

Resolution evidence: Production visual-map work after Phase B contracts load.

OF-010 - Full-game scale ceilings for markets, people, promotions and Chronicle bytes/year.

Source: GDD v0.2 §5.3; Technical Architecture v0.2 §17 Q9

Current contract: Schemas contain no scenario-count ceilings.

Resolution evidence: Soak tests and target-device profiling.

OF-011 - How much simulation detail is retained for minor off-screen entities at global scale.

Source: Technical Architecture v0.2 §17 Q14

Current contract: Same identity/reference contracts apply; detail-reduction policy is outside schema v0.1.

Resolution evidence: Global-scale performance evidence.

OF-012 - Ownership succession serialization beyond the current owner/promoter seat.

Source: GDD v0.2 §3.3 and §31 Q7; Technical Architecture v0.2 §17 Q16

Current contract: OwnershipSeatState identifies current promotion and owner with transition_pending extension point; no dynasty/shareholder succession mechanics invented.

Resolution evidence: Succession feature design.

OF-013 - Exact numeric formulas, heat decay, breakout incidence and victory thresholds.

Source: GDD v0.2 Document Control: exact numerical tuning OPEN; §31 Q3-Q4/Q12

Current contract: State uses normalized representational domains; tuning thresholds/formulas live in ruleset data and are not embedded in entity schemas.

Resolution evidence: Simulation and playtesting.

OF-014 - Future community mod signing/dependency tooling and executable extension policy.

Source: Technical Architecture v0.2 §5.4 and §17 Q17

Current contract: v0.1 accepts validated data/assets only; executable references are forbidden.

Resolution evidence: Post-slice security/compatibility design.

13. Phase 0 Exit-Gate Audit

Build-plan exit criterion	Status	Evidence
Every slice-critical entity has unambiguous identity and ownership/reference model.	PASS	Sections 1-2; schemas and semantic reference validator.

Build-plan exit criterion	Status	Evidence
Static definition data and mutable campaign state are distinguishable.	PASS	Section 3; ContentRegistry vs CampaignState matrix.
No Great Lakes, 1975, three-promotion, or population ceiling is hard-coded.	PASS	Generic ID/count schemas; no maxItems ceilings on campaign entity collections; campaign/date/map are content refs.
Knowledge projection boundary represented before UI.	PASS	KnowledgeBase/KnowledgeProjection schemas; hidden leak fixture rejected.
NarrativeContext boundary represented before content/UI.	PASS	Three-value enum + surface compatibility semantic validation.
Chronicle boundary represented before implementation.	PASS	Separate Chronicle manifest/events/metrics/projections/checkpoints/deltas/index/artifact/tombstone contracts.
Save version/migration boundary represented.	PASS	State/save/Chronicle schema versions, content fingerprint, transactional migration contract.
Package precise enough for Phase A-C without screens inventing domain shape.	PASS	Machine schemas, wrappers, fixtures and validator provide code-generation target.
Open questions recorded instead of guessed.	PASS	Open-Field Ledger section and JSON ledger.

Phase 0 conclusion: EXIT GATE SATISFIED. The next controlled engineering step is Phase A repository/project skeleton and headless boot, with Phase B campaign/content skeleton able to proceed in parallel once both consume these contracts. Any Phase A-C implementation that changes identity, reference, command, hidden-information, save, or Chronicle semantics must be treated as architecture-sensitive rework rather than a local convenience.

Appendix A. Field Catalog for Slice-Critical State

PersonState

Field	Type	Presence	Notes
id	RuntimeId	required	
seed_definition_id	nullable/union	required	
lifecycle	string	required	
legal_name	string	required	
display_name	string	required	
ring_name	nullable/union	required	
birth_date	Date	required	
pronoun_set_id	ContentId	required	
home_region_id	nullable/union	required	
tag_ids	array	required	
skills	object	required	
traits	object	required	
potential	object	required	
roles	array	required	
health	HealthState	required	
relationship_ids	array	required	
audience_by_market	object	required	
career	CareerState	required	
active_contract_ids	array	required	
current_hot_state	nullable/union	required	

PromotionState

Field	Type	Presence	Notes
id	RuntimeId	required	
seed_definition_id	nullable/union	required	
lifecycle	string	required	
brand_name	string	required	
controlling_owner_person_id	nullable/union	required	
ownership_model_id	nullable/union	required	
ownership_shares	array	required	
cash	Money	required	
prestige	NormalizedScalar	required	
locker_room_morale	NormalizedScalar	required	
momentum	SignedUnitScalar	required	
home_market_ids	array	required	
contract_ids	array	required	
touring_company_ids	array	required	
championship_ids	array	required	
program_ids	array	required	
media_deal_ids	array	required	
agreement_ids	array	required	
strategy_profile_id	nullable/union	required	
active_effect_ids	array	required	
current_hot_state	nullable/union	required	

MarketState

Field	Type	Presence	Notes
market_id	ContentId	required	
wrestling_interest	NormalizedScalar	required	
economic_strength	NormalizedScalar	required	
influence_by_promotion	object	required	
current_hot_state	nullable/union	required	
active_effect_ids	array	required	

RegionState

Field	Type	Presence	Notes
region_id	ContentId	required	
current_hot_state	nullable/union	required	
active_effect_ids	array	required	

TouringCompanyState

Field	Type	Presence	Notes
id	RuntimeId	required	
promotion_id	RuntimeId	required	
name	string	required	
person_assignment_ids	array	required	
carried_championship_ids	array	required	
route	array	required	
directives	array	required	

Field	Type	Presence	Notes
monthly_budget	Money	required	
fatigue_pressure	NormalizedScalar	required	
cohesion	NormalizedScalar	required	
current_hot_state	nullable/union	required	
status	string	required	

ContractState

Field	Type	Presence	Notes
id	RuntimeId	required	
person_id	RuntimeId	required	
promotion_id	RuntimeId	required	
status	string	required	
start_date	Date	required	
end_date	nullable/union	required	
compensation	ContractCompensation	required	
exclusivity_id	ContentId	required	
clause_ids	array	required	
clause_parameters	object	required	
leverage	NormalizedScalar	required	
dispute_state_id	nullable/union	required	

ChampionshipState

Field	Type	Presence	Notes
id	RuntimeId	required	
promotion_id	RuntimeId	required	
definition_id	ContentId	required	
status	string	required	
prestige	NormalizedScalar	required	
holder_person_ids	array	required	
recognition_market_ids	array	required	
recognition_promotion_ids	array	required	
last_change_event_id	nullable/union	required	

ProgramState

Field	Type	Presence	Notes
id	RuntimeId	required	
promotion_id	RuntimeId	required	
status	string	required	
started_on	Date	required	
ended_on	nullable/union	required	
side_a	ProgramSide	required	
side_b	ProgramSide	required	
purpose_id	ContentId	required	
phase_id	ContentId	required	
heat	NormalizedScalar	required	
momentum	SignedUnitScalar	required	
objective_ids	array	required	
planned_direction_id	nullable/union	required	

Field	Type	Presence	Notes
chemistry_truth	nullable/union	required	
current_hot_state	nullable/union	required	

MediaDealState

Field	Type	Presence	Notes
id	RuntimeId	required	
promotion_id	RuntimeId	required	
medium_id	ContentId	required	
outlet_id	ContentId	required	
status	string	required	
reach_market_ids	array	required	
start_date	Date	required	
end_date	nullable/union	required	
schedule_id	ContentId	required	
cost	nullable/union	required	
revenue	nullable/union	required	
exclusivity_id	nullable/union	required	
production_requirement_ids	array	required	
restriction_ids	array	required	

VenueState

Field	Type	Presence	Notes
venue_id	ContentId	required	
availability_state	string	required	
relationship_by_promotion	object	required	
active_restriction_ids	array	required	

AgreementState

Field	Type	Presence	Notes
id	RuntimeId	required	
party_promotion_ids	array	required	
status	string	required	
start_date	Date	required	
end_date	nullable/union	required	
clause_ids	array	required	
clauses	array	required	
trust_effect	SignedUnitScalar	required	
last_violation_event_id	nullable/union	required	

KnowledgeBase

Field	Type	Presence	Notes
owner_promotion_id	RuntimeId	required	
observations	array	required	
familiarity_by_subject	object	required	

Appendix B. Canonical Validation/Error Codes

Code	Meaning
ID001	Malformed canonical ID.
ID002	Duplicate/conflicting identity.
REF001	Missing required reference.
REF002	Collection key/canonical ID mismatch.
REF003	Reference targets wrong family/type.
STATE001	Value violates representational range/type.
STATE002	Duplicate/incompatible active assignment.
STATE003	Domain invariant violation.
CMD001	Unknown/invalid command type or payload.
CMD002	Command stale for current turn/state.
KNOW001	Player-facing projection attempts to expose hidden truth outside knowledge contract.
NAR001	Narrative-bearing content missing/invalid NarrativeContext.
NAR002	NarrativeContext not accepted by target surface.
MOD001	Unsafe pack path/path traversal/remote reference.
MOD002	Executable content reference forbidden by data-only mod policy.
SAVE001	Content fingerprint or required pack mismatch.
SAVE002	Unsupported newer save/schema version.
CHR001	Chronicle event/head/sequence identity invalid.
CHR002	Historical entity reference cannot resolve.
CHR003	Checkpoint + delta reconstruction mismatch.
RNG001	Simulation randomness bypassed RandomService (test/lint/runtime diagnostic).

Appendix C. Source-to-Contract Traceability

Contract area	Primary governing source
CampaignState vs ChronicleStore	Technical Architecture v0.2 §4.1, §7.12, Appendix D; GDD v0.2 §21.
Shared Person and role assignments	GDD v0.2 §8; Technical Architecture v0.2 §4.2-4.3; ADR-004.
Markets/influence/formal territory	GDD v0.2 §§5-6; Technical Architecture v0.2 §4.5, §7.8; Architecture Constitution 4-5.
Touring companies	GDD v0.2 §11; Technical Architecture v0.2 §4.6.
Commands/turn events	Technical Architecture v0.2 §6; Build Plan v0.1 §5.0.4.
Randomness	Technical Architecture v0.2 §6.4; ADR-009.
Static content/modding	GDD v0.2 §26-27; Technical Architecture v0.2 §5; ADR-005/011.
Knowledge boundary	GDD v0.2 §16; Technical Architecture v0.2 §7.10; ADR-007.
NarrativeContext/kayfabe	GDD v0.2 §§2.7, 20.5; Technical Architecture v0.2 §10.3; ADR-019.
Save/migration/recovery	Technical Architecture v0.2 §8; ADR-010.
Chronicle reconstruction/tombstones	GDD v0.2 §21; Technical Architecture v0.2 §7.12/§8.6; ADR-015 through ADR-018.
Phase 0 scope/exit gate	Development & Build Plan v0.1 §5.